

From push to position

A cross-app review of Iranian Android updates around the 2026 war: delivery paths, location telemetry and the evidence needed to connect them.

DIVAR / BALAD / NESHAN / SNAPP / TAPSI

14 MAY

Divar chain observed

2023

24 FEB

Divar chain returns

2026

28 FEB

War begins

2026

These are dated file observations and an external war marker. They do not establish deployment, a shared operator or a delivered second stage.

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Why these apps matter

A signed update can change how an app receives commands or sends observations without asking for a new Android permission.

In [my Divar post](#), I argued that a push-gated loader could make a popular app a route to a targeted implant or exploit on the phone. [The original code notes](#) describe the inserted class. I expanded the review to navigation and ride-hailing apps because those apps have legitimate reasons to see where devices travel. A collection path in one app and a delivery path in another would be a consequential combination. The APKs reviewed here do not show that combination being used.

LONGER HISTORY

Divar's loader predates this war

Sampled Divar **11.8.1** already carried the push-to-stage chain (VT: 14 May 2023). It is absent in sampled **11.14.9** and returns in **11.14.10** (VT: 24 Feb 2026). Neshan's separate pull route appears 19 Feb.

FROM JUNE

Location paths expand

Balad **4.81.1** and Snapp driver **5.17.0** add server-gated radio/location paths. Tapsi **7.8.4** has a geolocation POST; Neshan **14.10.2.3** adds opaque native HTTPS, later accepting GPS-containing input. Traffic is unverified.

CROSS-APP VERDICT

No malicious collection shown

The Balad, Snapp and Tapsi radio/location paths are compatible with ordinary positioning or signal checks. No attacker control, targeting, upload receipt or shared operator is verified. Neshan's silent diagnostic and native paths remain unresolved.

How to read the dates

A VT first-submission timestamp shows when that exact hash was in VT. An archive listing is weaker evidence of availability. Neither dates development, rollout, installation or use. In the charts, circles mark sampled code; vertical phase bands are external chronology, not attribution.

Scope: Divar, Balad, Neshan, Snapp and Tapsi; passenger and driver packages are kept distinct.

Divar's older loader

Exact APKs put the same push-to-stage workflow in 2023. The later points show sampled removals and returns, not continuous deployment.

SOLID = CHAIN PRESENT

HOLLOW = CHAIN ABSENT

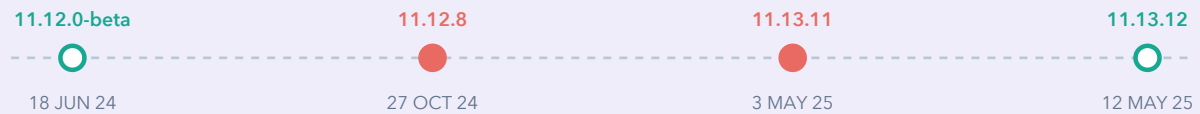
Dashes show order, not continuity

2023 / FIRST OBSERVED



Gaza fighting ended 13 May; the 11.8.1 file reached VT on 14 May. No operational link is known.

2024-25 / MIXED SAMPLES



The June negative is a beta flavor. 11.12.8 reached VT 27 Oct, after 26 Oct strikes; its build code suggests 20 Oct.

2026 / RETURN, INTERRUPTION, RETURN



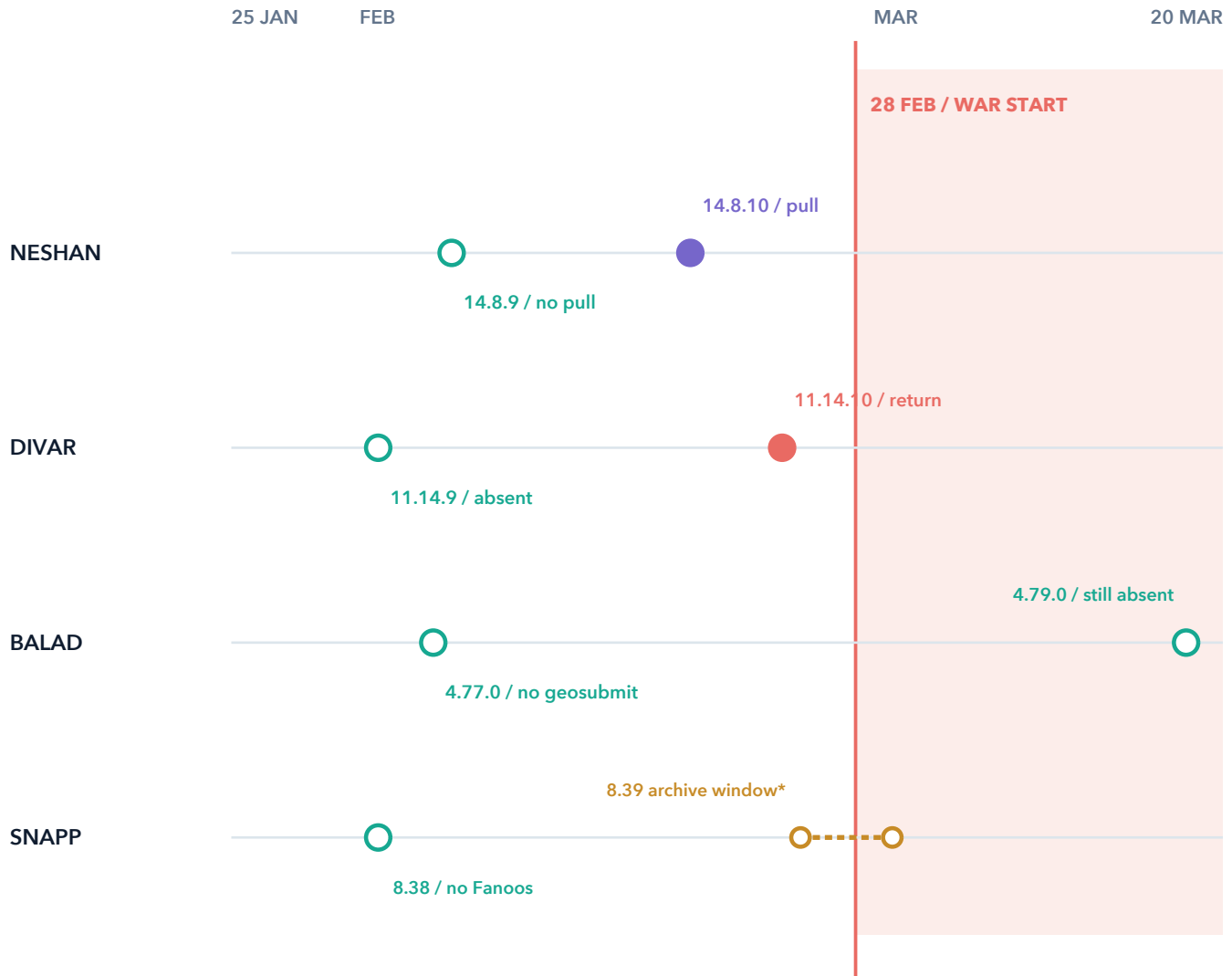
The 24 Feb file predates the 28 Feb war; a verified attack push or second stage remains absent.

What survived, what changed

The core fetch, decode/write, reflective invoke and cleanup sequence recurs. The 2023 URL is plain **callback_url**; later samples use encoded **campaign** (XOR 63, then 104). The worker retains a 14-field User-Agent after the app reaches 16 fields. These details support code lineage, not operator attribution.

The February reappearance

The meaningful overlap is an observed file window. It is not a record of coordinated targeting.



Pre-war files

Neshan's new pull route and Divar's returned push chain were in VT before the strikes. Neither point proves targeting during the war.

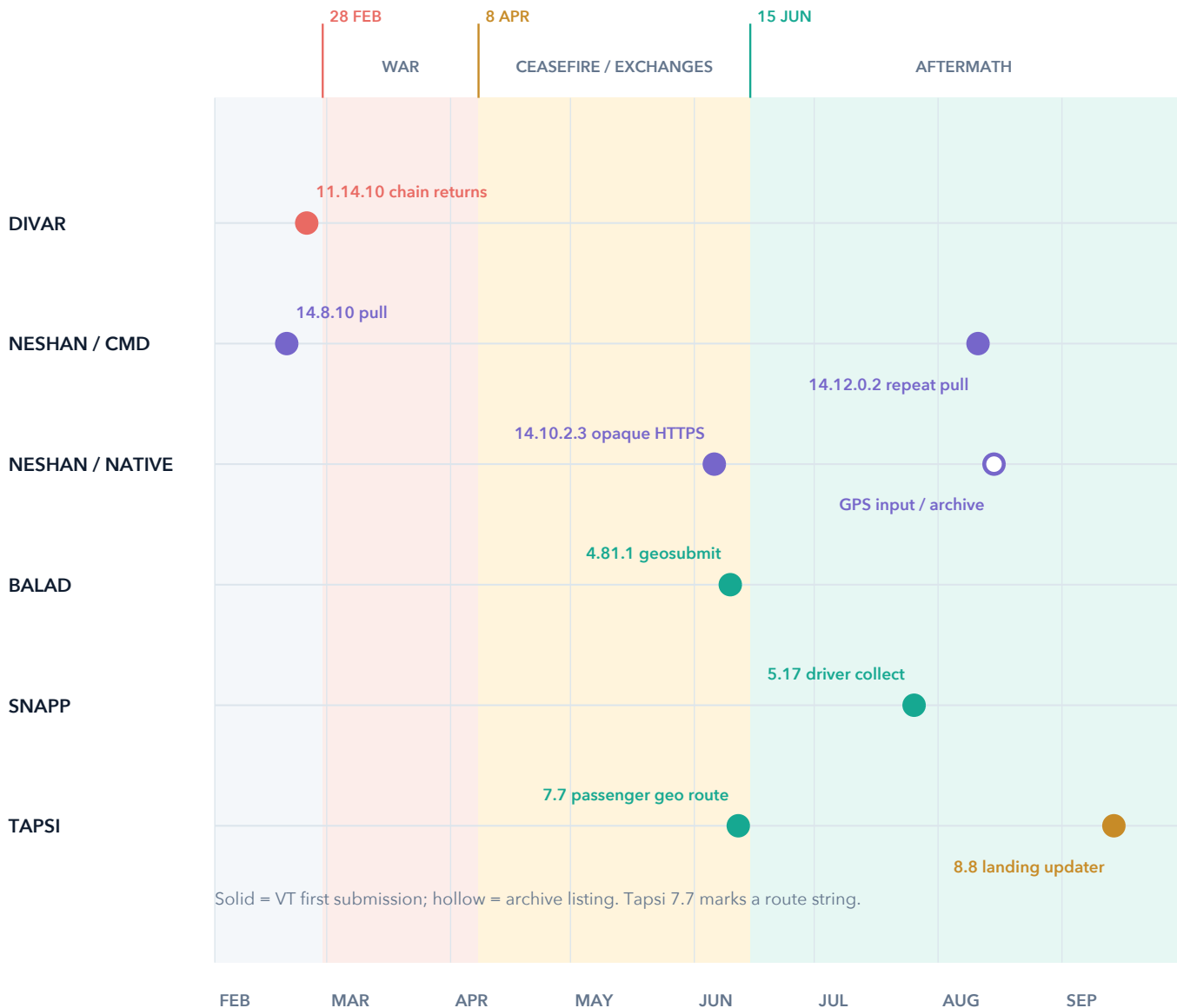
BadeSaba context

Reuters reported hostile BadeSaba app messages during the strikes. That incident does not identify the mechanism or actor behind the sampled Divar code.

Chart source: VT for app-code checkpoints; amber span = version-level archive listings; UN and Reuters for context.

How the paths changed

Phase bands track public war milestones. They should not be read as uninterrupted fighting or evidence of intent.



The timing that matters

Balad geosubmit, Neshan Impulse, Tapsi geolocation and Snapp radio collection first appear in sampled files from June onward. The April ceasefire period included intermittent exchanges. Missing intermediate APKs can move an introduction earlier, but no acquired file dates these specific collection paths to pre-war positioning. The overlay invites log recovery; it is not a causal finding.

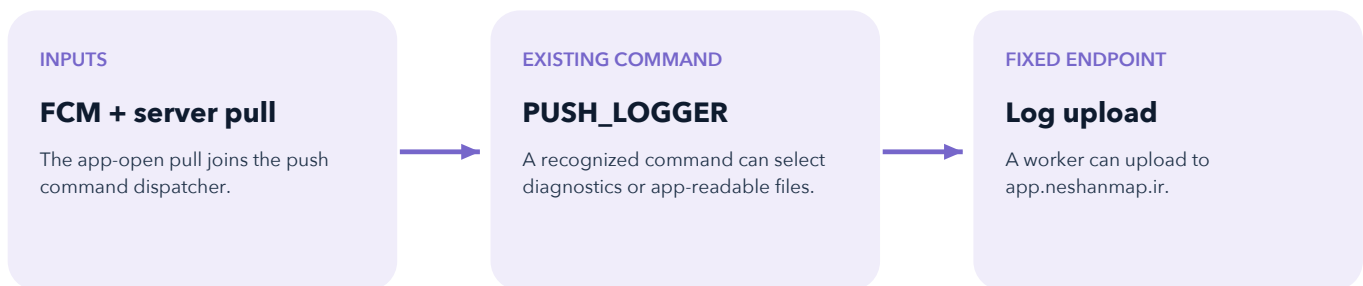
Delivery and collection stay separate

Solid arrows are paths established in sampled code. The dotted join below is a hypothesis that needs operational evidence.

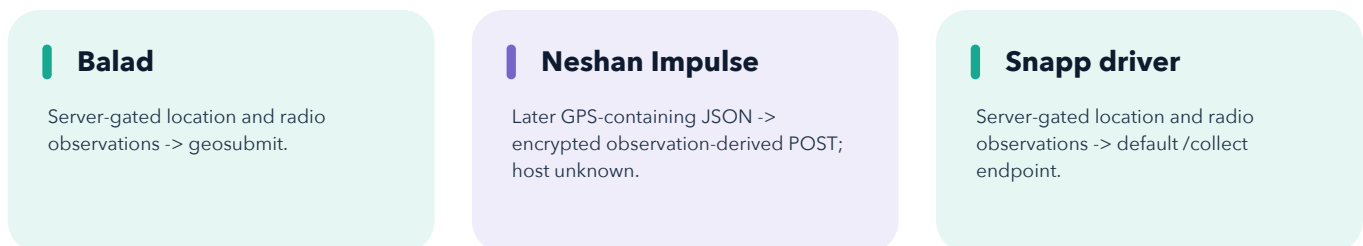
● DIVAR / TARGETED DELIVERY



● NESHAN / COMMAND AND DIAGNOSTICS



● SEPARATE LOCATION PATHS / SELECTED



Tapsi's positioning POST to its production API is reviewed on page 8.

HYPOTHESIS ONLY

If one operator controlled a location-data source and Divar push targeting, observations from one app could inform whom to send a stage to in another. There is **no verified common device join, shared operator, target list, push receipt or stage** in this record.

Off-band means a separate HTTPS path here; it does not itself mean covert or unauthorized.

Balad and Neshan

The review looked beyond Divar-shaped loaders, including off-band HTTPS and data-bearing native paths.

Balad: a server-gated radio uploader

The Bazaar **ir.balad** track is negative for the reviewed uploader in sampled **4.79.0** (VT: 18 Mar). Sampled **4.81.1** (VT: 10 Jun) calls **RadioObservationUploader** from map/navigation location updates. When **isGeoSubmitEnabled** is true, code accepts non-mock GPS/fused fixes, rate-limits them, and builds an observation from location and nearby cell/Wi-Fi readings for POST to **location.raah.ir/v2/geosubmit**. The model also contains a hash of a persistent app UUID and trip/map context. Later **4.84.0** can batch up to ten observations and use a separate network-location source switch.

The client defaults the master switch to false if absent. Some model fields are transient; exact wire fields, historical configuration, upload receipts and endpoint ownership remain unverified. The uploader is not called by the inspected FCM handler. Fine/coarse location, Wi-Fi-state and Internet permissions were already present before this route; the relevant delta is code and server control, not a new permission prompt.

Neshan: more ways to reach PUSH_LOGGER

The diagnostics/file-upload command predates the war in a sampled November 2025 APK. In **14.8.10** (VT: 19 Feb), a new **/notification/pull** request passes returned content to the same dispatcher as FCM. A recognized **PUSH_LOGGER** instruction can run without posting a user-visible notification and upload selected app-readable material to **app.neshanmap.ir/logger/uploadLogFile**. Sampled **14.12.0.2** adds a repeat pull worker. A static traversal concern could broaden selected files within the app's readable scope. No server response or upload receipt was recovered.

Neshan: native HTTPS evolves separately

Sampled **14.10.2.3** (VT: 6 Jun) introduces **Impulse** JNI code with an opaque native HTTPS request path. Its **nativeGetAll** wrapper takes no Java observation string, so later GPS-derived behavior cannot be read back into June. By sampled **14.13.0.6**, the revised method accepts Java observation JSON that can include the last GPS fix, incorporates input-derived data and constructs an encrypted JSON POST. The host/path is encoded; plaintext destination, exact transmitted fields, activation and historical traffic are unresolved. No call from the push/pull dispatcher to Impulse was found.

Source: preserved APK delta notes and VT timestamps in this evidence package.

Snapp and Tapsi

Passenger and driver packages are analyzed as separate products; a valid signature is only a lineage check.

Snapp: new driver radio collection path

In the 17 acquired Snapp APKs, driver **5.13.0** and **5.15.0** lack the reviewed collection path. Both sampled **5.17.0** variants (first VT: 26 Jul / 4 Aug) and **5.18.0** have it. A server AB flag, count and interval settings gate a POST combining precise location, nearby Wi-Fi/cell observations, ride state and time. The default endpoint is **locations.snapp.site/v1/driver/collect**; a server base URL can replace it. Location and Wi-Fi permissions were already present. Configuration and traffic were not recovered.

Passenger **8.39.0** adds a Fanoos notification poll, but its version archives disagree across 25 Feb / 2 Mar and the preserved APK came from the March listing. Its reviewed renderer posts visible, tap-driven notifications; fresh-install activation in that passenger build is unverified. The driver poll path is present by sampled **5.15.0**. Neither is a demonstrated Divar-style silent command loader. Same-version install-permission variants also carry a Behrooz in-app updater; no attack URL or stage was recovered.

Tapsi: geolocation RPC and an update flavor

The 33 acquired current-package APKs begin in June for passengers and July for drivers; January-May is uncovered. A sampled passenger **7.7.0** has the geolocation route by 12 Jun. The reviewed passenger **7.8.4** and driver **8.8.0** landing build have callable paths to POST cell/Wi-Fi observations and prior positions to **/v3/geolocation/geolocate**, which returns a location estimate. Production bases are **p1.tapsi.ir** and **d1.tapsi.ir**. This is a sensitive data-transfer capability in an apparent first-party positioning service; no historical traffic or non-Tapsi recipient was observed.

Two signed driver **8.8.0** files differ by distribution flavor. The September landing build includes Yadegar: it requests a server-selected patch URL from **d1.tapsi.ir**, downloads/applies a binary patch and invokes Android PackageInstaller. Its Myket counterpart lacks that module. The landing manifest adds install/update permissions. No separate expected hash, detached signature or pinned patch host was found in the inspected Java path; OS installer checks still apply. No attacker patch or malicious install was recovered.

Why these are bounded findings

No exact Divar class or calibrated relay/worker rule matched the acquired Snapp or Tapsi APKs. That check cannot clear missing releases or server-driven behavior. Tapsi passenger 7.8.4 and 7.10.0 use the same seven-field geolocation request; renamed classes are not a new upload. Its 8.8.0 difference is a distribution flavor, not proof of insertion and retraction. Backend configuration and device traffic would decide historical use.

Versions, permissions and blind spots

The table summarizes behavior changes, not a verdict on whether a company or signer authorized them.

APP / TRACK	ACQUIRED BOUND	RELEVANT DELTA	WHAT IS MISSING
Divar	11.14.9 -> 11.14.10	Push-to-stage chain returns	Attack push; stage; rollout
Balad / Bazaar	4.79.0 -> 4.81.1	Geosubmit appears	4.78.x; 4.80/4.81.0; config
Neshan	14.8.9 -> 14.8.10	Silent pull added	Pull responses; uploads
Neshan	14.10.2.3 -> 14.13.0.6	Observation-derived POST	Destination; wire fields
Snapp / driver	5.15.0 -> 5.17.0	Radio / location collect path	5.16.0; config; traffic
Tapsi / p + d	7.7.0 / 7.21.0	Geolocation route present	Jan-May; config; traffic

Permissions can stay still while behavior changes

Balad, Neshan and Snapp driver already had relevant location/network permissions at the reviewed collection boundaries. A new route need not prompt a new grant. Tapsi's landing-flavor updater instead adds install/update declarations. Runtime grants, foreground state, OS installer checks and actual traffic remain device-specific.

Signer continuity does not clear an update

The certificate groups related APKs; it cannot tell whether a build was authorized. A stolen key or compromised release process can sign malicious code. Every version delta here is assessed on code and provenance, not on signature validity alone.

Records that would settle the question

Correlate on exact devices, times and operator-controlled infrastructure. Overlapping file dates alone are not enough.

01

Build and distribution history

Developer CI attestations, signed build inputs, store/CDN rollout cohorts, and retained APKs for all gaps. Compare release-branch approval against exact hashes, including same-version variants.

02

Delivery and command logs

Divar OneSignal/FCM sends, payload bodies and target subscriptions; Neshan FCM and /notification/pull responses, PUSH_LOGGER dispatch and upload receipts; Snapp Fanoos polls, flags and device notification histories. Check bodies that never became visible notifications.

03

Location and network receipts

Balad AppMeta flags and geosubmit receipts; Neshan impulseConfig and a controlled native-request trace; Snapp isWifiCollectionEnabled, dynamic base URL and /collect receipts; Tapsi geolocate requests, feature flags, Yadegar patch manifests and install events.

04

Cross-app join test

Only attempt device/person joins with independently verified account identifiers, device-side consented telemetry or lawful backend records. A hashed app-local UUID is not shown to be a cross-app identifier.

SELECTED SOURCES

[My Divar post](#)[UN May 2023 Gaza](#)[UN war start](#)[UN June peace deal](#)[Divar code notes](#)[UN Oct 2024 Iran strikes](#)[UN April ceasefire](#)[Reuters on BadeSaba](#)

Companion evidence: divar_2023_lead/, snapp/, tapsi/ and timeline/events.csv.

Files and endpoints to verify

These are code-level pivots from relevant changed paths. A host, hash or class here is not a malicious verdict by itself.

EXACT SAMPLED FILES

DIVAR	11.8.1	14 May 2023 / VT
ee36859f872cf5cf3723f0cfde50adf5c61409a273ff268d6d0660f83c326a81		
NESHAN	14.8.10	19 Feb 2026 / VT
947c4a9412ef577b29c87d87764b4e9e0e3ddfd1966bff2cbab83adb3f2f8115		
DIVAR	11.14.10	24 Feb 2026 / VT
9e6a9955d9b6e1db5636d4213c386944da8e740170f4e6159b2ecb202decdb0db		
NESHAN	14.10.2.3	6 Jun / VT
90499f420b9288d8574e9bf9ae01ee90b2dc1dabb3005877b8655468c204a698		
BALAD	4.81.1	10 Jun / VT
da7e348ba2fd161e0a0d05b7ad7a216f95389f9e69219144d5cc72bf7454370b		
TAPSI PASSENGER	7.8.4	19 Jul / VT
e7e7597fcc8e2dc93535bd8eebd862433da34c1c67a1c73120e3268a84ae2869		
SNAPP DRIVER	5.17.0	26 Jul / VT
2c7bee97b304134c58aafc15d9935a55a9091488e2a200080353218932bfe671		

CODE-LEVEL NETWORK PIVOTS

BALAD	location.raah.ir/v2/geosubmit
	Server-gated location/radio observations
NESHAN	app.neshanmap.ir/notification/pull
	Command source joined to FCM dispatcher
NESHAN	app.neshanmap.ir/logger/uploadLogFile
	Existing PUSH_LOGGER upload destination
TAPSI	p1.tapsi.ir / d1.tapsi.ir
	POST /api/v3/geolocation/geolocate; positioning service
SNAPP DRIVER	locations.snapp.site/v1/driver/collect
	Default base URL; server-gated radio collection
TAPSI UPDATER	d1.tapsi.ir/api/v3/yadegar/public/get-patch
	Server returns patchUrl; landing flavor only
NESHAN IMPULSE	plaintext host unresolved
	Encoded native configuration; do not invent an IOC

Reproduce the boundary

The companion CSVs retain full hashes and date types. The app folders hold acquired files and methods. Repeat both the positive-path review and the earlier negative checks before making a release-wide claim.